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How generative AI reshapes gratitude interventions: Benefits and risks

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ABSTRACT

As AI-supported mental health resources proliferate, understanding how Generative AI (GenAI) reshapes the process of well-being interventions—not just their outcomes—is critical. We conducted a 7-day randomized controlled trial comparing the Gratitude Chatbot condition (GenAI-mediated conversational journaling) against the Gratitude Journal condition (structured-prompt journaling) and an active Control condition (daily events writing). Linear mixed-effects model results revealed that, compared with the Journal condition, participants in the Chatbot condition reported less perceived effort and expressed shallower gratitude. Chatbot interactions also shifted the orientation of gratitude, eliciting more self-directed expressions. Both gratitude conditions enhanced happiness, meaningfulness, and positive linguistic expression relative to the active control, with no significant differences observed between the two gratitude formats. Thematic analysis complemented these findings, showing that participants reported feeling surprised and inspired by the GenAI-generated summaries. Our findings reveal GenAI-mediated conversational journaling as a double-edged sword: it reduces cognitive barriers, fosters inspiration, and promotes self-oriented gratitude, but may simultaneously erode reflective depth. These findings inform the design of future GenAI-supported well-being technologies that weigh the benefits and costs of AI mediation, allowing individuals to decide whether GenAI integration suits their personal needs.

1. Introduction

1.1. *Integrating generative AI into gratitude intervention*

Global declines in life satisfaction and rising emotional distress, particularly among younger cohorts (Helliwell et al., 2024; Piao et al., 2024), necessitate scalable, low-barrier mental health interventions. Gratitude journaling is a well-established practice for enhancing two distinct forms of well-being (Bolier et al., 2013). The first is hedonic well-being, which refers to the experiential dimension of psychological flourishing, encompassing positive affect, happiness, and the absence of negative affect (Diener, 1984). A one-week gratitude intervention has been shown to increase happiness and reduce negative affect one month later (Seligman et al., 2005). The second is eudaimonic well-being, a meaning-centered dimension that emphasizes personal growth and purpose in life (Ryan & Deci, 2001; Ryff, 1989). Gratitude practice has similarly been found to be positively associated with eudaimonic well-being (Nezlek et al., 2017). However, high cognitive costs often hinder the efficacy of gratitude journaling. Sustained engagement requires deliberate identification of gratitude targets and deep emotional reflection—demands that frequently lead to diminished adherence among stressed or low-motivation populations (Emmons & McCullough, 2003).

The emergence of Generative AI (GenAI) offers a transformative solution. Beyond serving as delivery platforms, GenAI systems (e.g., Rosebud AI, Myentries.ai) can interactively prompt and elaborate on user entries by synthesizing contextually nuanced responses (Sengar et al., 2025). However, current research has predominantly focused on outcome effectiveness—whether AI-assisted tools improve well-being or reduce depression, while neglecting the fundamental question: How does GenAI reshape the psychological trajectory of the gratitude practice itself? This study argues that GenAI is not a neutral conduit; rather, it redistributes effort investment, alters both the targets and depth of gratitude, and recalibrates state well-being and reflective insight, presenting a nuanced landscape of psychological risks and opportunities.

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1.2. *Cognitive offloading: altered effort and gratitude depth*

From a process-oriented perspective, cognitive offloading—the use of external tools to reduce internal mental demands (Risko & Gilbert, 2016)—provides a critical framework for understanding GenAI's impact. While offloading to AI can lower the barrier to daily journaling by externalizing processing demands, this efficiency may incur a cognitive tax. A classic cognitive offloading study documented by Sparrow et al. (2011) demonstrated that when individuals encounter difficult questions, they immediately think of search engines such as Google or Yahoo. Furthermore, when people know that information is readily accessible online, they are less likely to encode the information itself and instead tend to remember where to retrieve it. This tendency to outsource cognitive labor to external search tools has been termed *the Google effect*. Building on this foundation, Kosmyna et al. (2025) found that participants who relied on AI for writing tasks over an extended period exhibited poorer memory retrieval performance and reduced brain connectivity compared to those who did not rely on any digital tools. These findings suggest that habitual AI reliance may erode the depth of cognitive engagement over time.

According to Lin's (2015) hierarchical model of gratitude, the psychological benefits of gratitude practice are not uniform across levels of cognitive engagement but rather increase progressively with deeper elaboration. The model delineates four ascending levels of gratitude processing: (1) recognizing kindness—the most basic acknowledgment of a positive event; (2) feeling grateful—the internalization of an affective response; (3) expressing appreciation—the active communication of gratitude toward a benefactor; and (4) returning favor—the behavioral manifestation of reciprocal prosocial motivation. Empirical evidence supports this hierarchy: deeper levels of elaboration are associated with greater positive affect, stronger social bonding, and more sustained well-being gains (Emmons & McCullough, 2003; Regan, Walsh, & Lyubomirsky, 2023). These findings align with broader cognitive theories suggesting that generative processing—actively constructing meaning from experience rather than passively receiving it—is the primary mechanism through which emotional benefits accrue (Craik & Lockhart, 1972). Drawing on cognitive offloading theory, we thus investigate whether GenAI-assisted gratitude journaling reduces perceived effort while simultaneously attenuating the depth of gratitude processing.

1.3. *Reshaping gratitude targets: the digital mirror effect*

Beyond effort, GenAI may pivot the thematic focus of gratitude. Traditionally, gratitude is viewed as an interpersonal construct directed toward external benefactors (Lambert et al., 2009). However, GenAI often functions as a *digital mirror*, facilitating self-referential reflection and metacognitive awareness (Tomisu et al., 2025). Evidence suggests that AI interaction can disproportionately heighten self-awareness (Li et al., 2025), potentially shifting the target from other-oriented appreciation to self-oriented gratitude (e.g., acknowledging personal resilience). In this light, GenAI-mediated conversational journaling, compared to structured-prompt journaling, may offer a novel pathway toward self-kindness, a dimension that often remains secondary in conventional gratitude practices.

1.4. *Affective alignment and integrative feedback*

GenAI may also modulate state well-being through affective alignment. Through digital emotional contagion, users may subconsciously synchronize their mood with the affirming, positively valenced language typical of AI chatbots (Han et al., 2023; Neumann & Strack, 2000). This process is theoretically grounded in emotional contagion theory (Hatfield et al., 1993), which posits that individuals automatically and continuously mimic and synchronize with the emotional expressions of others—a mechanism originally documented in face-to-face interaction but subsequently extended to text-based and computer-mediated communication (Cheshin et al., 2011; Kramer et al., 2014). Consistent with these mechanisms, Pataranutaporn et al. (2023) found that the linguistic output of generative AI was more positively valenced than that of human interlocutors, and that as conversations progressed, participants who held more positive beliefs about the AI exhibited increasingly positive language patterns. We therefore hypothesize that the positive linguistic valence characteristic of GenAI will promote individuals' state well-being—indexed by momentary happiness and sense of meaning in life—as well as the positivity of their own linguistic expression.

Crucially, GenAI's most potent contribution may lie in post-hoc integration. Unlike isolated daily reflections, GenAI can synthesize longitudinal entries into a coherent narrative, identifying latent themes that elude individual perception (Blyler & Seligman, 2024). By reorganizing fragmented thoughts into higher-order representations (Wright et al., 2026), GenAI-generated feedback may catalyze novel insights. This study explores the perceived accuracy and surprise factor of such AI-driven synthesis in fostering deeper self-reflection.

1.5. *The present study*

Adopting a process-oriented lens (Fig. 1), this study examines how GenAI-mediated conversational journaling, relative to structured-prompt journaling, reconfigures the gratitude intervention across four dimensions:

Hypothesis 1. (Cognitive Effort): GenAI-mediated conversational journaling will attenuate participants' perceived cognitive effort during

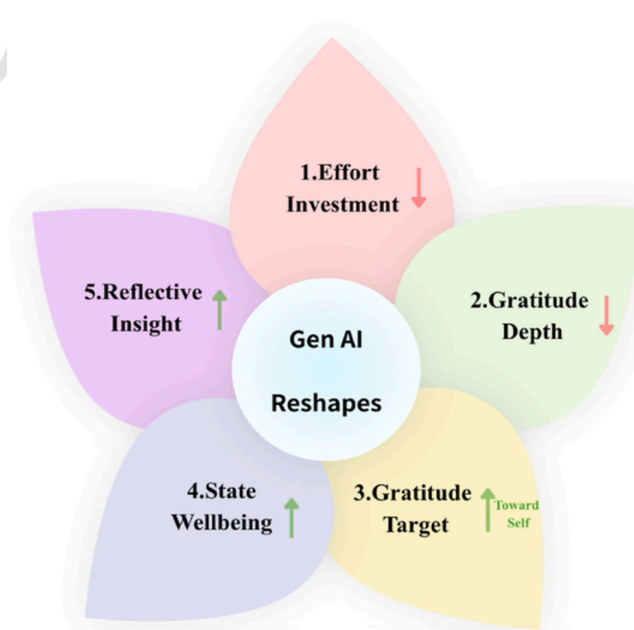

Fig. 1. The five ways that GenAI may reshape the process of gratitude interventions

Note. Each petal represents a dimension through which GenAI-mediated conversational journaling may reshape gratitude journaling practice, relative to structured-prompt journaling. Specifically, GenAI-mediated conversational journaling may reduce participants' degree of psychological effort investment (Hypothesis 1); reduce gratitude depth (Hypothesis 2) while potentially promoting more self-oriented gratitude expressions (Hypothesis 3); enhance state well-being outcomes (Hypothesis 4); and promote personal reflective insight through consolidated GenAI-generated feedback (Hypothesis 5).

journaling, relative to structured-prompt journaling.

Hypothesis 2. (Gratitude Depth): GenAI-mediated conversational journaling will lead to a decrease in the depth of gratitude processing, relative to structured-prompt journaling.

Hypothesis 3. (Gratitude Target): GenAI-mediated conversational journaling will promote self-oriented gratitude, relative to structured-prompt journaling.

Hypothesis 4. (State Well-being): GenAI-mediated conversational journaling will enhance state well-being relative to structured-prompt journaling via positive affective alignment.

Hypothesis 5. (Reflective Insight): GenAI's integrative feedback will cumulatively deepen user insight and self-awareness.

2. Methods

2.1. Participants

The sample size was determined via a power analysis for linear mixed-effects models using GLIMMPSE (Kreidler et al., 2013). Based on an estimated effect size of $d = 0.558$ from prior research on robot-expressed gratitude (Lee et al., 2024), a minimum of 102 participants (34 per group) was required to achieve a power of 0.80 ($\alpha = .05$). A total of 141 Taiwanese adults were recruited online in July 2025, of whom 111 completed the baseline assessment and commenced the intervention. Six participants were subsequently excluded for completing fewer than five entries, yielding a final analytical sample of 105 (M age = 23.38, range: 18–56; 63.8% female). Attrition did not differ significantly across conditions (Fisher's exact test, $p = .489$; more detail see Supplementary Material A). Regarding educational attainment, 70.7% held bachelor's degrees and 20.7% held graduate degrees. Prior chatbot use varied considerably, with most participants reporting frequent use (one to three times per week or more; 76.2%), while a small minority reported rarely or never using chatbots (11.4%).

2.2. Procedure and design

A mixed-design was employed with three conditions: Gratitude Chatbot (n = 36), Gratitude Journal (n = 36), and a Daily Events Control (n = 33). Following baseline demographic collection, participants were randomly assigned to their respective groups for a 7-day intervention (Fig. 2).

- The GenAI-mediated conversational journaling Condition (i.e., *Chatbot*): Participants interacted with a GPT-4-powered chatbot designed with a "mindful, lively, and genuine college student" persona. This persona was chosen to enhance interpersonal communication and foster gratitude (Sawyer et al., 2022; Zeng et al., 2025). Each session involved identifying three grateful events, with a limit of three conversational turns per event to ensure consistent interaction depth (Fig. 3). For detailed chatbot prompt design and procedural specifications, please refer to Supplementary Material B.
- The structured-prompt journaling Condition (i.e., *Journal*): Participants recorded three grateful events daily via an online survey platform using sequential prompts, mimicking traditional journaling (Cunha et al., 2019).
- The events writing condition (i.e., *Control*): Participants factually recorded three daily events without emotional reflection or gratitude-specific prompts.

All entries were completed between 19:00 and 24:00 (UTC + 8) to ensure temporal consistency. Following the 7-day period, participants in the three conditions received AI-generated integrative feedback synthesized from their collective entries and evaluated its impact on self-reflection.

2.3. Material

2.3.1. Gratitude and state well-being

To verify the manipulation and assess outcomes, we administered the following (all on a 6-point Likert scale):

- Gratitude: Assessed via gratitude awareness (adapted from GQ-6; McCullough et al., 2002) and willingness to express gratitude (Lambert et al., 2010).
- Perceived effort: Given that longitudinal studies inherently impose a certain degree of participant burden (Eisele et al., 2022), we extracted the most relevant item from the NASA-TLX (Hart & Staveland, 1988) to capture participants' perceived effort during gratitude journaling, thereby minimizing response fatigue.
- State well-being: Measured through hedonic well-being (momentary happiness; Lyubomirsky & Lepper, 1999) and eudaimonic well-being (presence of meaning; Steger et al., 2006).

2.3.2. AI-generated integrative feedback

Following the review of AI feedback, participants evaluated four dimensions: (1) alignment with lived experience (accuracy), (2) perceived surprise, (3) self-awareness gains, and (4) novel insights. These items were adapted from Blyler and Seligman (2024), supplemented by open-ended questions for qualitative depth. The complete AI feedback setting prompt and examples are provided in Supplementary Material C.

2.4. Data analysis

2.4.1. Quantitative analysis

Linear mixed-effects models with random intercepts were used to analyze the longitudinal questionnaire data. To account for potential baseline differences, gender, age, and prior chatbot use frequency were included as covariates in all models, as previous research has suggested that they may influence either human-chatbot interactions or gratitude reflection (Chopik, Newton, Ryan, Kashdan, & Jarden, 2019; Kashdan et al., 2009; Law et al., 2023). Holm-adjusted pairwise comparisons based on estimated marginal means were conducted to control the family-wise error rate (Abdi, 2010)() and complete pairwise comparison tables are provided in Supplementary Material A (see Table A1).

To supplement self-reports, the emotional valence of entries was analyzed using SnowNLP (Zhang and Zhang, 2022) , a Python library for Chinese sentiment analysis. Higher scores (0–1) indicated greater positive valence (). To evaluate the validity of SnowNLP for sentiment scoring in the context of gratitude journaling, a random sample of 90 entries was independently rated by human coders and two large language models (Rathje et al., 2024). Both human and LLM ratings showed significant positive correlations with SnowNLP scores ($ps < 0.05$), supporting its validity in gratitude context. More detailed validation results are provided in Supplementary Material D.

2.4.2. Qualitative analysis

2.4.2.1. Deductive content analysis. Drawing on Lin's (2015) hierarchical framework, we coded Gratitude Depth as a four-level hierarchy of increasing cognitive elaboration: (1) Recognizing kindness (e.g., " No one was pushing or shoving on the subway."), (2) Feeling grateful (e.g., "I found a Japanese dessert I had bought a long time ago and tried it—it was surprisingly delicious! ... I was overjoyed!"), (3) Expressing appreciation (e.g., "Thank you, Abao, for spending the whole day with me."), (4) Returning favor (e.g., "The pork set lunch was delicious... I thanked the staff on the spot and left a tip"). Gratitude Target coding was adapted from Adler and Fagley (2005), categorizing entries into

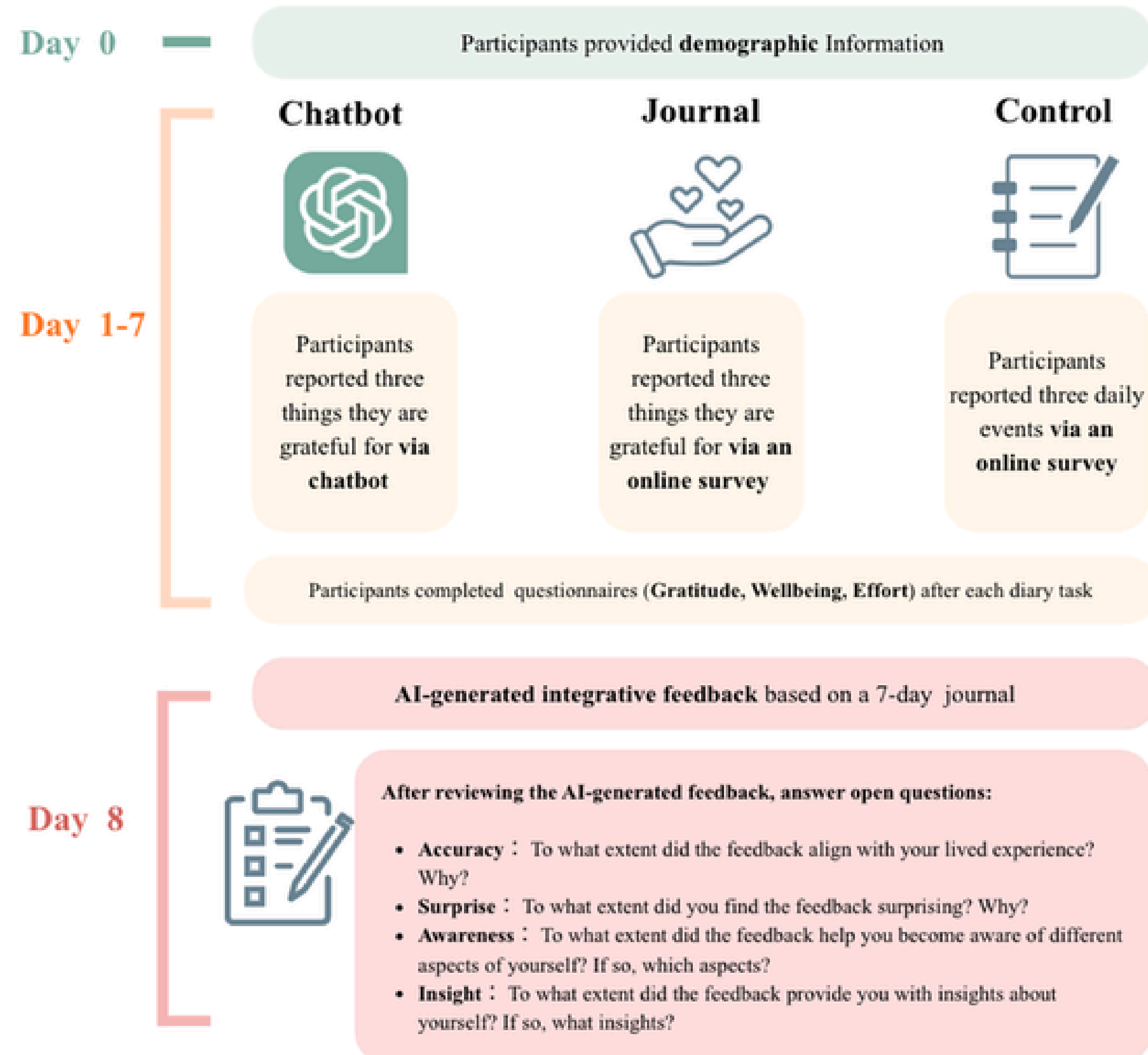

Fig. 2. Study procedure of the 7-day intervention among the Chatbot, Journal, and Control conditions
Note. Only participants in the Chatbot condition interacted with a chatbot. Those in the Journal and Control conditions logged gratitude or daily event entries on Qualtrics.

three domains: Self-oriented (e.g., "The power went out at my rental place, but thankfully it came back on after 30 min"), Other-oriented (e.g., "Thank you to my friend for celebrating my birthday with me today."), and Environment-oriented (e.g., "The scenery was beautiful. The buildings near campus were stunning.") The complete coding book and examples is provided in Supplementary Material E.

The content analysis sample was restricted to participants in the Chatbot and Journal conditions, as the Control condition did not involve gratitude writing entries and was not included in the qualitative coding. In addition, to minimize the potential influence of condition awareness on coding decisions, only user-generated text was submitted for content analysis; AI-generated chatbot turns were removed from the Chatbot condition transcripts prior to coding. Coders were instructed to evaluate entries based solely on their semantic content. Interrater reliability for Gratitude Depth and Gratitude Target, assessed across two independent coders, was 0.93 and 0.90, respectively.

Chi-square tests were used to compare categorical distributions between the Chatbot and Journal conditions. Since multiple coded entries were nested within participants, additional generalized linear mixed-effects models with participant-level random intercepts were conducted to account for within-person clustering. Separate binomial models were estimated for each gratitude depth and gratitude target category using dummy-coded outcomes. In addition to gender, age, and chatbot use frequency, word count was also included as a covariate to account for variation in response length, as prior study has suggested that word count may affect gratitude reflection (Regan et al., 2023).

2.4.2.2. Inductive thematic analysis. Two participants in the Chatbot condition dropped out at this stage, resulting in 103 participants who provided complete qualitative responses to the AI-generated integrative feedback. Participants' open-ended responses to the AI-generated integrative feedback were analyzed using thematic analysis (Braun & Clarke, 2006). Each response was treated as an independent unit of analysis. In the initial coding phase, meaningful semantic segments were identified and labeled with descriptive codes that closely reflected the original text. Codes capturing similar psychological meanings were then grouped and consolidated into subthemes. Finally, subthemes sharing a common underlying psychological response were merged into higher-order themes. This process yielded 24 subthemes organized under four overarching themes, each representing a distinct type of expe-

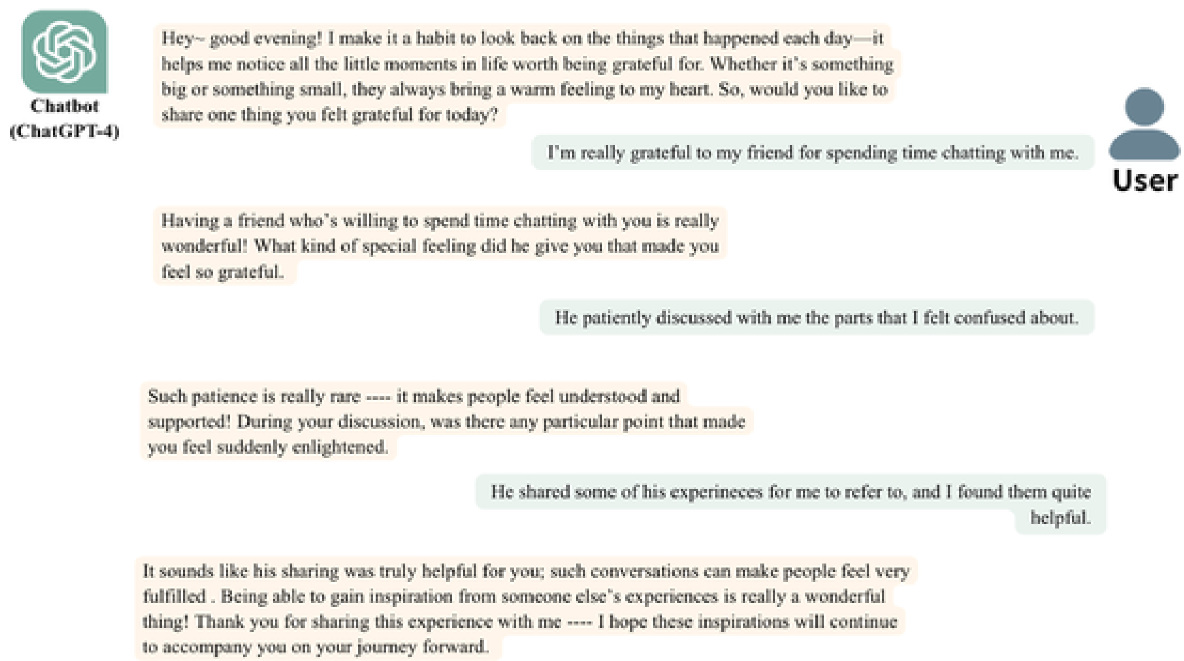

Fig. 3. Example of chatbot-user interaction in the Chatbot condition
Note. This figure presents a representative interaction between a participant and the AI chatbot (ChatGPT-4) in the Chatbot condition. During the study, participants engaged with the interface entirely in Chinese; the English text shown here is a translation provided for clarity.

riential response to AI-generated narrative feedback: Accuracy, Surprise, Awareness, and Insight.

2.5. Transparency in ethics and data handling

This study was approved by the Research Ethics Committee (NTU-REC-202401HS030). Prior to enrollment, all participants provided written informed consent and were explicitly informed that they would interact with an AI-powered chatbot. Each journaling session was processed independently, with no conversational context retained across sessions. For the post-intervention feedback, participant responses were processed via GPT-4. Before processing, all entries were manually reviewed by the research team, and any personally identifiable information was removed.

This study was not preregistered. Hypothesis 1 (perceived effort) and 4 (state well-being) were fully theory-driven and confirmatory in nature. Hypothesis 2 (gratitude depth) and 3 (gratitude target) were confirmatory with respect to the predicted direction of group differences; however, the specific operationalization and categorization involved some refinements to the coding scheme during analysis. Hypothesis 5 (post-intervention feedback) was fully exploratory in nature and was not specified in advance. To enhance research transparency, all data and codes are available on the Open Science Framework(https://osf.io/fn3tj/overview?view_only=5de485844db5403fb5997543321b91ae).

3. Results

3.1. Manipulation check

Linear mixed-effects models confirmed the effectiveness of the gratitude intervention. Compared to the Control group, participants in both the Chatbot ($b = 0.42$, $p = .060$, marginal significant) and Journal conditions ($b = 0.46$, $p = .049$) reported higher gratitude awareness. Similarly, both gratitude groups exhibited a greater willingness to express gratitude relative to the Control group (Chatbot: b = 0.56, $p = .042$; Journal: b = 0.47, $p = .067$, marginal significant). No significant differences were found between the Chatbot and Journal conditions for either awareness ($b = 0.04$, $p = .852$) or willingness ($b = 0.09$, $p = .689$), indicating that both formats successfully elicited gratitude.

3.2. GenAI's impact on effort and gratitude depth

To test Hypothesis 1 (Effort Investment), we compared perceived effort across conditions. A theoretically meaningful pattern emerged when both gratitude conditions were compared to the Control group: the Journal condition required significantly more effort than the Control ($b = -0.62$, $p = .025$), whereas the Chatbot condition did not significantly differ from the Control ($b = 0.38$, $p = .229$). Nevertheless, the direct contrast between the Chatbot and Journal conditions did not reach statistical significance ($b = -0.25$, $p = .293$), indicating that the evidence for attenuated effort in chatbot-guided journaling relative to structured-prompt journaling is modest. This pattern partially suggests that GenAI-mediated conversational journaling may attenuate the perceived cognitive demands of gratitude reflection to a level more comparable to neutral event recording, though this interpretation should be treated with caution given that the two journaling conditions did not differ significantly from each other.

Regarding Hypothesis 2 (Gratitude Depth), a chi-square test revealed significant differences between the Chatbot and Journal conditions ($\chi^2(3, 72) = 16.55$, $p < .001$; Fig. 4). Post-hoc analysis (Fig. 4) showed that the Chatbot condition produced significantly more content at the most basic level (Recognizing Kindness; $z = 3.93$, $p < .001$), whereas the Journal condition yielded more content at the feeling grateful ($z = -2.23$, $p = .026$) and expressing appreciation ($z = -2.75$, $p = .006$) levels. No significant difference was observed between the two gratitude conditions for returning favor ($z = 0.76$, $p = .446$). To improve statistical stability, feeling grateful and expressing appreciation were merged into a deep gratitude category, while recognizing kindness served as shallow gratitude; Returning Favor was excluded due to insufficient observations. Age, gender, prior chatbot ex-

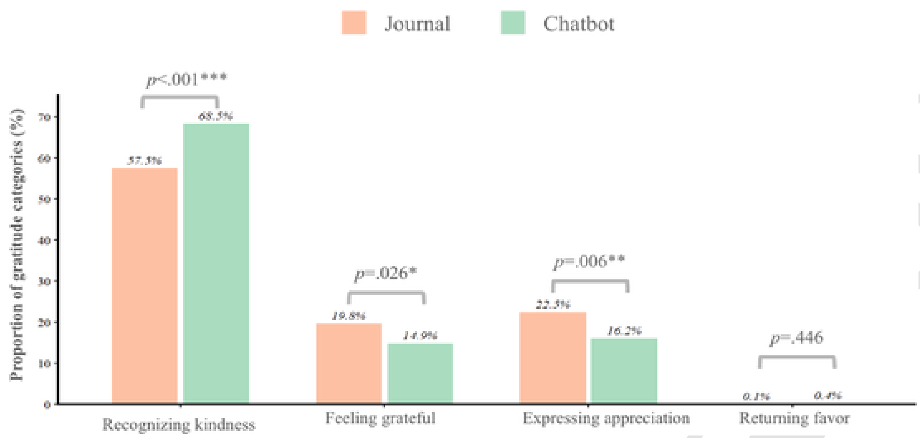

Fig. 4. The distribution of gratitude-depth categories in the Journal and Chatbot conditions
Note. The orange bars represent the percentage of responses at each gratitude depth level in the Journal condition; the green bars represent the percentage of responses at each gratitude depth level in the Chatbot condition. The statistical notations indicate the results of pairwise comparisons following a chi-square test. $*p < .05$, $**p < .01$, $***p < .001$; N = 72.

perience, and word count were included as covariates. Generalized linear mixed-effects models revealed that participants in the Chatbot condition were more likely to remain at the shallow acknowledgment level of gratitude ($b = -1.21$, $p = .018$), whereas participants in the Journal condition were more likely to express deeper gratitude, including feeling grateful and expressing appreciation ($b = 1.21$, $p = .018$). These results suggest that while AI attenuates participants' perceived cognitive effort, it may lead to shallower emotional processing.

3.3. Reshaping gratitude targets: the shift toward self

In support of Hypothesis 3 (Gratitude Target), the focus of gratitude significantly differed between groups ($\chi^2(2, 72) = 9.89$, $p = .007$; Fig. 5). Post-hoc comparisons indicated that participants in the Chatbot condition expressed significantly more self-oriented gratitude than those in the Journal condition ($z = 3.14$, $p = .002$; Fig. 5). Conversely, participants in the Journal condition expressed significantly more other-oriented gratitude than those in the Chatbot condition ($z = -2.24$, $p = .025$). No significant differences were found for environment-oriented gratitude ($z = -0.84$, $p = .402$). To verify the robustness of these findings, a generalized linear mixed-effects model with participant-level random intercepts was employed, with age, gender, prior chatbot experience, and word count included as covariates. Results indicated that only self-oriented gratitude remained significant, with Chatbot condition participants being more likely to express self-oriented gratitude ($b = -0.54$, $p = .040$). Differences in other-oriented gratitude were no longer statistically significant ($b = 0.53$, $p = .117$), and environment-oriented gratitude remained non-significant ($b = 0.01$, $p = .982$). Taken together, the most robust finding is that GenAI shifts the reflective focus toward the user's personal agency and internal states.

3.4. State well-being and affective alignment

To test Hypothesis 4 (State well-being), linear mixed-effects models examined happiness, meaningfulness, and text sentiment (Fig. 6). Both gratitude conditions—regardless of format—significantly outperformed the Control group in enhancing happiness (Chatbot: $b = 0.56$,

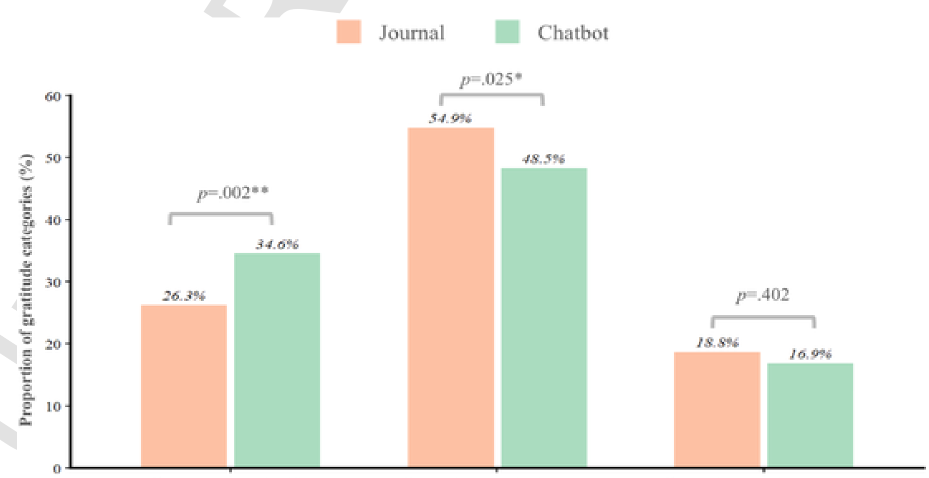

Fig. 5. The distribution of gratitude targets showing standardized residuals in the Journal and Chatbot conditions.
Note. The orange bars represent the percentage of specific gratitude targets in the Journal condition; the green bars represent the percentage of specific gratitude targets in the Chatbot condition. The statistical notations indicate the results of pairwise comparisons following a chi-square test. $*p < .05$, $**p < .01$, $***p < .001$ N = 72.

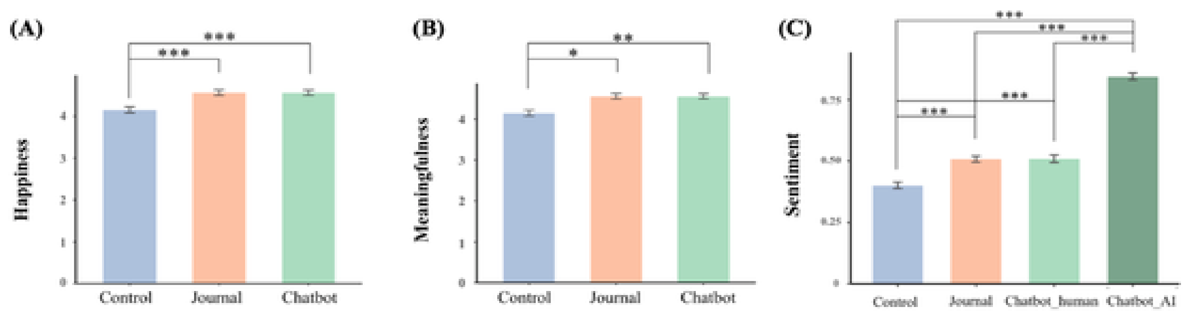

Fig. 6. Mean state well-being outcomes under different interventions

Note. The figure shows mean (A) happiness, (B) meaningfulness, and (C) sentiment of the three conditions (Control, Journal, and Chatbot). The Chatbot condition is further divided into "Chatbot_human" (participants' own responses during the conversation) and "Chatbot_AI" (the chatbot's responses) in (C). Happiness and meaningfulness represent hedonic and eudaimonic well-being, respectively, after journaling. Sentiment uses SnowNLP to analyze the positive emotion score in the text: the higher the score, the more positive the text content. *$p < .05$, **$p < .01$, ***$p < .001$. N = 105.

$p = .012$; Journal: $b = 0.61$, $p = .005$) and meaningfulness (Chatbot: $b = 0.46$, $p = .049$; Journal: $b = 0.47$, $p = .049$), suggesting that the practice of gratitude itself promotes state well-being independent of delivery format. No significant differences were observed between the Chatbot and Journal conditions for happiness ($b = -0.05$, $p = .791$) or meaningfulness ($b = -0.01$, $p = .954$), indicating that GenAI-mediated conversational journaling did not confer incremental state well-being benefits beyond those achievable through structured-prompt journaling alone. Notably, although SnowNLP analysis confirmed that AI-generated language exhibited significantly higher positive sentiment than human-generated text across all three conditions (*ps* < 0.001; the rightmost bar in Fig. 6C), this did not translate into elevated emotional states among participants in the Chatbot condition, whose sentiment scores were nearly identical to those in the Journal condition ($b = 0.01$, $p = .688$). These findings suggest that the positive affective alignment mechanism proposed in H4 was not operative in the present study.

3.5. Exploratory analysis: insights from GenAI feedback

To test Hypothesis 5 (Reflective Insight), ANOVA results indicated no significant differences in how participants across the three conditions perceived the AI-generated feedback (Accuracy ($F(2, 100) = 0.50$, $p = .611$), Surprise ($F(2, 100) = 1.84$, $p = .165$), Awareness ($F(2, 100) = 1.51$, $p = .225$), and Insight ($F(2, 100) = 1.32$, $p = .271$)), suggesting the benefits of integrative feedback are robust across journaling formats (Fig. 7).

Consolidated qualitative analysis identified four core themes from participants' responses to the AI-generated integrative feedback. Following the quantitative distributions, qualitative responses rated at moderate levels or above were thematically analyzed across four dimensions—Accuracy, Surprise, Awareness, and Insight—yielding a series of subthemes that were subsequently merged into four overarching themes (see Table 1 for subthemes and representative quotes):

- The first theme, Touching Moments in Routine Life (Accuracy), captures participants' recognition that the feedback accurately reflected their lived experience through precise attunement to the texture of their daily lives—including their self-concept, values, and habitual patterns—rather than through abstract generalization.
- The second theme, Event Integration Capability (Surprise), characterizes participants' astonishment at the AI's capacity to synthesize fragmented daily entries into coherent, higher-order self-narratives, revealing dimensions of self-understanding that had remained inaccessible through individual reflection alone.
- The third theme, Perceiving Deeper-Level Needs (Awareness), reflects the heightened self-awareness participants reported following engagement with the feedback—a clarification of inner states, needs, and habitual patterns that had previously remained implicit or unacknowledged.
- The fourth theme, Cultivating Flourishing (Insight), represents the culminating psychological response, in which participants moved beyond recognition and awareness toward active meaning-making that generated forward momentum, reinforced positive self-regard, and translated into concrete intentions for self-care and growth.

4. Discussion

4.1. Efficiency vs. depth: the structural trade-off in AI-assisted reflection

Our results partially support Hypothesis 1, suggesting that GenAI-mediated conversational journaling may attenuate the perceived mental effort associated with gratitude reflection. Specifically, while structured-prompt journaling required significantly more effort than Control, the GenAI-mediated conversational journaling condition did not differ significantly from the Control. This pattern aligns with a cognitive offloading framework, wherein external tools reduce cognitive load (Peng & Yeh, 2025; Risko & Gilbert, 2016). Crucially, however, given that the direct contrast between the two journaling conditions was not statistically significant, the evidence that GenAI-mediated journaling reduces effort relative to traditional structured prompts remains modest. These findings should therefore be interpreted as suggestive rather than conclusive, underscoring the need for future research utilizing larger samples to rigorously validate this effect.

However, this reduction in "friction" comes at the cost of gratitude depth, supporting Hypothesis 2. This suggests that when GenAI handle the structural demands of a task, internal cognitive elaboration—critical for deep gratitude processing—may be truncated (Kosmyna et al., 2025; Mueller & Oppenheimer, 2014). We propose that this represents a structural trade-off: while GenAI lowers the entry barrier for populations with low motivation or high stress, it may simultaneously bypass the desirable difficulties (Bjork & Bjork, 2020) necessary for profound gratitude. Consequently, GenAI chatbots may be most effective as scaffolding for beginners, whereas traditional journaling remains the gold standard for seasoned practitioners seeking deeper level of reflection.

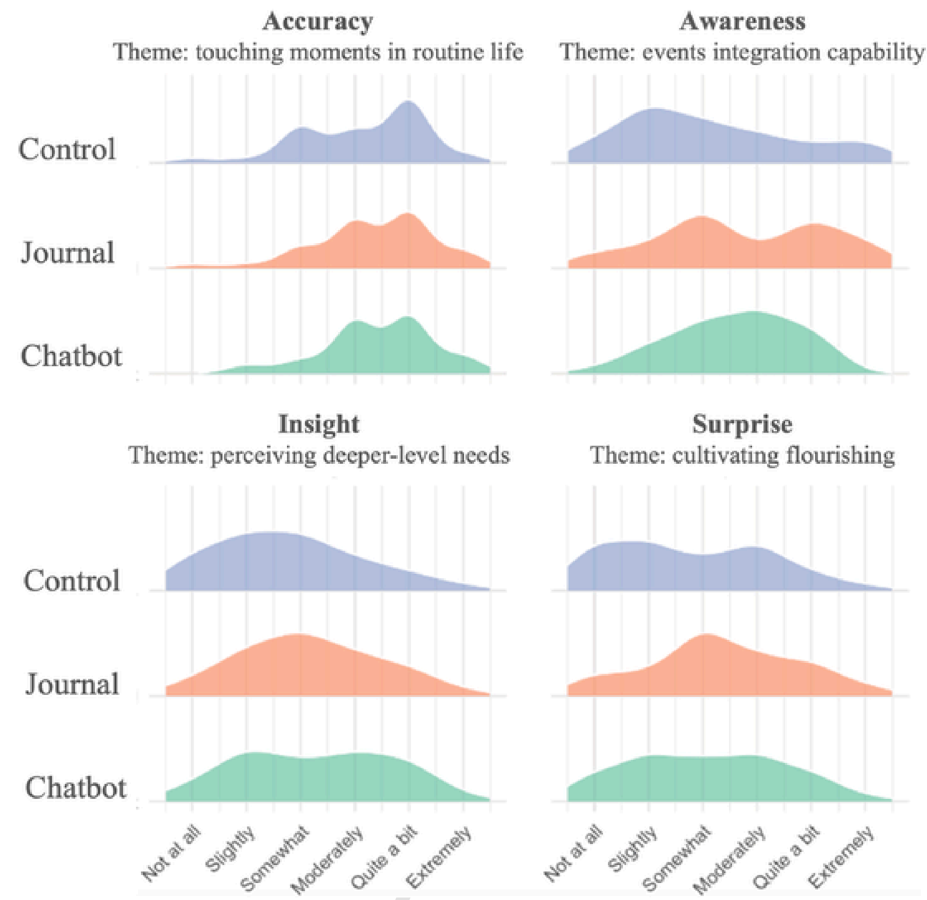

Fig. 7. Participants' Perceptions of AI-Generated Feedback after the 7-day interventions across three conditions
Note. The figure presents density plots for three conditions—Control (green), Journal (orange), and Chatbot (blue)—across four dimensions: accuracy, surprise, awareness, and insight. Two participants in the Chatbot condition did not complete this phase ($N = 103$).

4.2. GenAI as a digital mirror: the self-oriented shift

A pivotal finding was the significant increase in self-oriented (intrapersonal) gratitude in the Chatbot condition, supporting Hypothesis 3. Unlike structured-prompt journaling, which often emphasizes interpersonal benefactors (Lambert et al., 2009), the interactive nature of GenAI-mediated conversational journaling appears to function as a digital mirror—reflecting user input through personalized, affirmative dialogue and facilitating a reorientation of gratitude from a relational emotion toward a self-evaluative process. This shift manifested across three distinct intrapersonal sub-dimensions (Adler & Fagley, 2005), each illuminating a different psychological pathway through which GenAI may promote self-oriented flourishing.

The first pathway concerns have-type gratitude—noticing and appreciating what one already possesses. Participants in the Chatbot condition frequently acknowledged personal resources such as health and daily achievements (e.g., "I exercised on time today"). This resonates with the savoring literature, which identifies deliberate attention to positive experiences as a key mechanism for sustaining well-being (Bryant & Veroff, 2017; Jose et al., 2012). The second pathway concerns loss-type gratitude—reframing adversity as an occasion for appreciation. Participants in the Chatbot condition regularly reinterpreted negative experiences in a positive light (e.g., "It was pouring rain and I couldn't get a taxi, but luckily the bus sped up and I made it to my destination just in time.") This reappraisal process maps onto self-compassion, which emphasizes responding to difficulty with kindness and equanimity rather than self-criticism (Neff, 2003). The third pathway concerns comparison-type gratitude—generating positive affect by recognizing personal growth relative to one's past self. Participants in the Chatbot condition frequently reflected on progress and improvement (e.g., "Today I checked my grades and saw I got an A+ in an elective I really struggled with at the beginning — it feels like my effort paid off"). This orientation aligns with growth mindset theory (Dweck, 2016), which holds that individuals attuned to effort-based progress derive meaning from evidence of personal development.

Taken together, these three pathways suggest that GenAI's digital mirror effect operates as a multi-layered facilitation of intrapersonal flourishing—simultaneously scaffolding savoring of present resources, compassionate reframing of adversity, and recognition of personal growth. This positions GenAI-mediated conversational journaling as a promising platform for self-oriented well-being interventions, extending conventional gratitude practice into domains of self-evaluation that traditional formats have often left underexplored (Tachon et al., 2022).

4.3. The limits of digital affect: why emotional contagion fails

The results of Hypothesis 4 require careful interpretation. Both gratitude conditions significantly outperformed the Control in enhancing happiness and meaningfulness, confirming the well-established benefit of gratitude practice for state well-being. However, GenAI-mediated conversational journaling did not confer incremental well-being benefits beyond those of structured-prompt journaling, and participants did not mirror the chatbot's highly positive affective tone. These findings suggest that the observed well-being gains are attributable to gratitude

Table 1
Themes and subthemes identified in thematic analysis of post-intervention GenAI feedback.

Theme	% above threshold	Subthemes	Representative quote
1. Touching moments in routine life *(Accuracy)*	78.4%	1.1 Content fit with self-concept 1.2 Recognition of specific life details 1.3 Explanation of behavioral motives 1.4 Emotional needs identified by feedback 1.5 Order, control, and self-regulation noted	*"Indeed, I pay considerable attention to things that occur at regular intervals... this gives me a sense of stability and touches me."*
2. Event integration capability *(Surprise)*	40.1%	2.1 Discovery of unnoticed self-traits 2.2 Elevation of mundane to higher-order self-schema 2.3 Systematic narrative organization 2.4 Inner psychological capacities surfaced 2.5 Motivational structure made explicit	*"What surprised me most was... 'transformation from being supported to becoming a supporter'... I hadn't consciously noticed this while journaling."*
3. Perceiving deeper-level needs *(Awareness)*	47.0%	3.1 Awareness of gratitude frequency 3.2 Recognition of observational sensitivity 3.3 Identification of relational needs 3.4 Emotional need awareness 3.5 Order and control need recognition 3.6 Emotional regulation capacity identified	*"This description helped me better understand my desire for authentic relationships and self-growth."*
4. Cultivating flourishing *(Insight)*	37.2%	4.1 Gratitude reframed as abundant resource 4.2 Attitudinal reappraisal toward others 4.3 Heightened mindful attention to detail 4.4 Resilience and coping strategies affirmed 4.5 Goal-directed action and self-care intentions	*"It showed me that even amid a stressful life, I can still be grateful... It made me feel that I'm doing well."*

Note. % = percentage of participants whose ratings met or exceeded the moderate threshold and were therefore included in the thematic analysis. Themes are organized by coded category in parentheses. Representative quotes are translated from Traditional Chinese. Detailed descriptions of all subthemes, including theoretical rationale and representative participant quotes, are provided in Supplementary Material F.

journaling as a general practice (Bolier et al., 2013; Emmons & McCullough, 2003), rather than to the interactive or affective properties specific to GenAI-mediated journaling.

The most parsimonious explanation is that participants did not attribute sufficient emotional agency to the AI for affective contagion to occur. Emotional contagion is fundamentally a social phenomenon contingent on the perceiver's attribution of genuine mental states—including feelings, intentions, and consciousness—to the expressive agent (Hatfield et al., 1993). When interacting with AI systems, however, users frequently perceive expressed affect as algorithmically generated rather than genuinely felt, withholding the mental state attributions necessary to trigger contagion (Colombatto et al., 2025). This interpretation is further supported by research on the uncanny valley of mind (Stein & MacDorman, 2024), which suggests that entities perceived as lacking authentic inner experience elicit fundamentally different social-emotional responses than human interlocutors, even when their expressive output is formally identical. When the AI is not perceived as a genuine experiential agent, the mechanisms underlying affective contagion appear to be attenuated or bypassed entirely, leaving gratitude practice itself as the primary driver of well-being gains.

4.4. Integrative feedback: from daily log to life narrative

Our Hypothesis 5 highlights GenAI's unique strength in post-hoc integration. Regardless of the journaling format, 78.4% of participants found the AI-generated feedback accurate, with nearly half reporting enhanced self-awareness. By synthesizing fragmented entries into a coherent life narrative, GenAI helps users identify deeper-level needs and cultivate flourishing that are often obscured during daily recording (Heintzelman & King, 2019). This capacity to foster flourishing through nostalgia and synthesis (Abeyta & Pillarisetty, 2023) suggests that GenAI's most significant value may lie not in the moment of recording, but in the integration of long-term psychological data.

4.5. Limitations and future directions

This study has several limitations. First, a central interpretive limitation concerns the confounding between GenAI involvement and conversational format. The Chatbot condition differed from the Journal condition not only in the use of generative AI, but also in conversational turn-taking, adaptive prompt structure (Liu et al., 2022, pp. 4437–4452), chatbot persona, and perceived social presence (Nass & Moon, 2000; Reeves & Nass, 1996). Consequently, observed differences cannot be unambiguously attributed to GenAI per se. Future research should employ a more rigorous control design incorporating: (1) a scripted chatbot condition, isolating conversational format from AI-generated content, (2) a structured GenAI condition, isolating generative AI from multi-turn dialogue, or (3) a human conversational partner condition, to disentangle social presence from AI agency (Brandtzaeg and Følstad, 2017).

Second, perceived effort was assessed with a single item adapted from the NASA-TLX Effort dimension (Hart & Staveland, 1988), a decision necessitated by the longitudinal daily diary design in which repeated administration of the full instrument would have imposed excessive participant burden (Rhéaume & Mullen, 2018; Zeng et al., 2026). Future research should incorporate additional NASA-TLX dimensions to provide a more comprehensive assessment of how GenAI-mediated journaling affects the full spectrum of cognitive workload—such as Frustration to assess emotional resistance to the task (Hancock et al., 2013; Matthews et al., 2002), and Temporal Demand to examine perceived time pressure during reflection (Pickup et al., 2005).

Finally, to ensure equitable coding across conditions, AI-generated turns were removed prior to coding so that only user-generated text was analyzed. However, residual stylistic features of Chatbot entries may have remained recognizable to coders familiar with gratitude journaling, potentially introducing expectancy bias. Future research should adopt more stringent blinding protocols—ideally recruiting coders who were entirely blind to the study hypotheses and condition assignments (Losiewicz, Oard, & Kostoff, 2000; Yeaton & Sechrest, 1981). The most rigorous approach would involve standardizing entry format and length prior to coding, and having an independent third party assign randomized participant IDs before distributing materials to coders, thereby minimizing the risk of condition inference at every stage of the analytic process (Brewer & Nakamura, 1984; Kazdin, 2003).

5. Conclusion

This study demonstrates that integrating GenAI reconfigures the gratitude process across five distinct dimensions: (1) alleviating cognitive load, (2) attenuating reflection depth, (3) shifting focus toward self-oriented gratitude, (4) failing to produce incremental well-being gains beyond traditional gratitude journaling, with no evidence of emotional contagion from GenAI-generated language, and (5) facilitating individual inspiration and insight. The core challenge for future digital well-being design lies in achieving a human-AI symbiosis that leverages AI's narrative synthesis and accessibility while mitigating its tendency to bypass the effortful processing essential for sustaining long-term reflective depth.

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CRediT authorship contribution statement

Peng-Yu Zeng: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Kai-Yu Chang:** Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Kuan-Yu Lin:** Writing – original draft, Visualization, Software, Data curation. **Su-Ling Yeh:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Consent to participate

All participants provided written informed consent and agreed to data publication terms before participating in the study.

Ethics approval

This study was approved by the Institutional Review Board (NTU-REC-202401HS030). Before enrollment, participants provided informed consent and were explicitly informed that they would interact with an AI-powered chatbot and that each journaling session was independent, with no memory of prior interactions retained across sessions.

Data availability

All raw data and codes were stored in the OSF platform (https://osf.io/fn3tj/overview?view_only=5de485844db5403fb5997543321b91ae).

Declaration of generative AI and AI-assisted technologies in the writing process

To enhance clarity and readability, this study used ChatGPT to refine sentences. All content, ideas, analysis, and conclusions remain entirely the work of the authors.

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Declaration of competing interest

The authors have no conflicts of interest to disclose.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chbah.2026.100333.

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